STAT201 · STATISTICS · LECTURE 10

Hypothesis Testing I: Logic, the p-value and Errors

How to decide formally whether the data agree with a statement about a parameter, and what the researcher risks

Syllabus topic 10 · week 10 · 4 academic hours

LECTURE GOALS

- formulate the null and alternative hypotheses and choose a one-sided or two-sided test;
- carry out a z- and t-test for a mean and a z-test for a proportion in five steps;
- compute and correctly interpret the p-value;
- explain Type I and Type II errors, the significance level and power;
- relate hypothesis testing to confidence intervals and distinguish statistical significance from practical significance.

1 The logic of hypothesis testing

Hypothesis testing is a formalized answer to the question “do the data agree with a statement about the population?”. The scheme resembles the legal presumption of innocence: the null hypothesis is not rejected until the data provide sufficiently strong evidence against it.

HYPOTHESES

The *null hypothesis* H_0 is a statement of “no effect” (an equality sign): $\mu = \mu_0$, $p = p_0$. The *alternative hypothesis* H_1 (or H_a) is what we want to prove: two-sided $\mu \neq \mu_0$; one-sided $\mu > \mu_0$ or $\mu < \mu_0$. The direction is chosen *before* looking at the data, based on the research question.

EXAMPLE 1. FORMULATIONS

A bank promises to process an application in 12 days on average. Customers complain that it takes longer. $H_0 : \mu = 12$, $H_1 : \mu > 12$ (one-sided). If the manager wants to know whether the time has changed at all: $H_1 : \mu \neq 12$.

1.1 Five steps

1. Formulate H_0, H_1 and choose the significance level α (usually 0.05).
2. Check the conditions (randomness and independence, normality or large n).
3. Compute the test statistic.
4. Find the p-value (or the critical value).

5. Make a decision and state the conclusion in the language of the problem.

THE p-VALUE

The p-value is the probability (given that H_0 is true) of obtaining a value of the statistic as extreme as, or more extreme than, the observed one. A small p means: the observed result is hard to explain by chance under H_0. Decision: if $p \le \alpha$, H_0 is rejected; otherwise there is no basis to reject it.

COMMON MISTAKE

The p-value is *not* the probability that H_0 is true, and not the probability of an erroneous decision. It is $P(\text{data} \mid H_0)$, not $P(H_0 \mid \text{data})$ — the same transposition of conditions as in Lecture 3. Moreover, "we do not reject H_0" does not mean "we have proved H_0": perhaps there are too few data.

2 Tests for a mean

2.1 The z-test (σ known)

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}. \tag{2.1}$$

When H_0 is true the statistic is approximately $N(0,1)$. The p-value: two-sided $2P(Z > |z|)$; right-sided $P(Z > z)$; left-sided $P(Z < z)$. Critical values at $\alpha = 0.05$: ± 1.96 (two-sided), 1.645 (one-sided).

EXAMPLE 2. APPLICATION PROCESSING TIME, $\sigma = 4$, $n = 36$

$H_0 : \mu = 12$, $H_1 : \mu \neq 12$, $\alpha = 0.05$. (a) $\bar{x} = 12.4$: $z = \dfrac{0.4}{0.667} = 0.60$, $p = 2 \cdot 0.274 = 0.549 > 0.05$ — no grounds to reject H_0. (b) $\bar{x} = 13.6$: $z = \dfrac{1.6}{0.667} = 2.40$, $p = 2 \cdot 0.0082 = 0.0164 < 0.05$ — H_0 is rejected: the average time differs from 12 days.

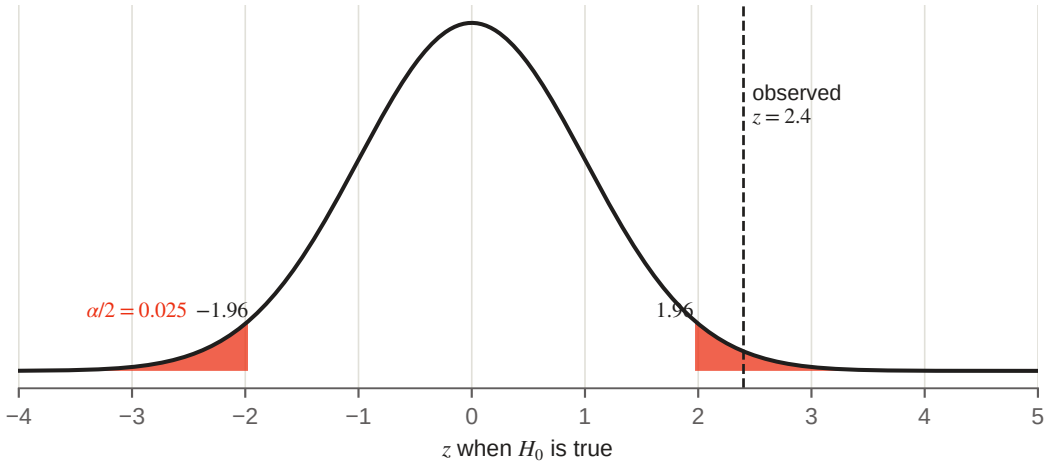


Fig. 1. A two-sided test at $\alpha = 0.05$: the rejection regions are 2.5% in each tail. The observed $z = 2.4$ falls into the right rejection region; the p-value is the area beyond $|z| = 2.4$ in both tails.

2.2 The t-test (σ unknown)

We replace σ by s and use the t-distribution with $n-1$ degrees of freedom:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1. \tag{2.2}$$

EXAMPLE 3. BRANCH REVENUE

The chain considers a monthly revenue of 30 million UZS per branch to be the norm. For 12 branches $\bar{x} = 35.75$, $s = 20.33$. $H_0 : \mu = 30$, $H_1 : \mu \neq 30$. $t = \dfrac{5.75}{20.33/\sqrt{12}} = \dfrac{5.75}{5.869} = 0.98$, $\nu = 11$, $p = 0.348$. The critical value is ± 2.201. H_0 cannot be rejected: the difference of 5.75 million is easily explained by chance with such a spread. Note that the 95% CI $(22.83;\ 48.67)$ contains 30.

EXAMPLE 4. A ONE-SIDED TEST

A sample $n = 25$, $\bar{x} = 48.2$, $s = 5$. $H_0 : \mu = 50$, $H_1 : \mu < 50$. $t = -1.8/1 = -1.80$, $\nu = 24$; $p = P(t_{24} < -1.80) = 0.042 < 0.05$ — we reject (critical value -1.711). A two-sided test would have given $p = 0.084$ and would not have rejected H_0: the direction must be chosen in advance and justified, not fitted to the result.

3 The test for a proportion

For $H_0 : p = p_0$ the standard error is computed *from* p_0 (we assume H_0 is true):

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \qquad \text{condition: } np_0 \geq 10,\ n(1-p_0) \geq 10. \tag{3.1}$$

EXAMPLE 5. DEFECTS ON A LINE

The permissible defect rate is $p_0 = 0.10$. In a sample of 200 items 30 are defective, $\hat{p} = 0.15$. $H_1 : p > 0.10$. $z = \dfrac{0.05}{\sqrt{0.09/200}} = \dfrac{0.05}{0.0212} = 2.36$, $p = 0.0092$. We reject H_0: the defect rate exceeds the permissible level.

EXAMPLE 6. SATISFACTION

The hypothesis “exactly 50% are satisfied” against $\neq$; 240 of 400 are satisfied. $z = \dfrac{0.10}{\sqrt{0.25/400}} = 4.00$, $p \approx 6 \cdot 10^{-5}$. We reject H_0. (In a CI for a proportion the standard error is computed from $\hat{p}$, in a test from p_0.)

4 Type I and Type II errors, significance level and power

	H_0 is true	H_0 is false
Do not reject H_0	correct $(1-\alpha)$	Type II error (β)
Reject H_0	Type I error (α)	correct: **power** $1-\beta$

The *significance level* $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$ is the risk of a false alarm; we set it ourselves. *Power* $1-\beta$ is the probability of detecting an effect when there is one. For a two-sided z-test at a true μ and $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$ the power equals $P(Z > 1.96 - \delta) + P(Z < -1.96 - \delta)$.

EXAMPLE 7. POWER

$H_0 : \mu = 12$, $\sigma = 4$, $n = 36$, $\alpha = 0.05$. If the true $\mu = 13.5$: $\delta = 2.25$, power $= 0.614$ (Type II error $\beta = 0.386$). At $\mu = 14$ the power is 0.851; at $\mu = 12.5$ only 0.117.

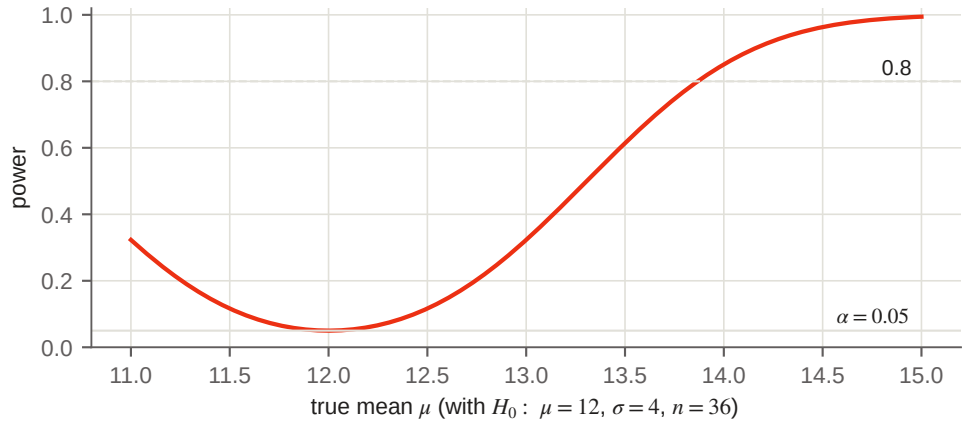


Fig. 2. The power of a two-sided z-test as a function of the true μ. At $\mu = 12$ (true H_0) it equals $\alpha = 0.05$; the larger the effect, the higher the power.

Power increases with n, with the size of the effect and with α, and with a decrease in σ. For 80% power and an effect $\Delta = \mu - \mu_0$ one needs $n \approx \left(\dfrac{(z_{\alpha/2} + z_\beta)\sigma}{\Delta} \right)^2$, where $z_\beta = 0.8416$. For $\Delta = 1.5$, $\sigma = 4$: $n = \left(\dfrac{2.80 \cdot 4}{1.5} \right)^2 = 55.8 \approx 56$.

REMEMBER

Lowering α reduces the risk of a false alarm but lowers power. To reduce both errors at once n must be increased. The choice of α depends on the cost of errors: for testing drug safety a false "approval" costs more than for a marketing hypothesis.

5 The link with confidence intervals

A two-sided test at level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 lies outside the $(1-\alpha)$ CI. In example (b): the 95% CI at $\bar{x} = 13.6$ is $(12.29;\ 14.91)$, 12 lies outside — the test rejects; for (a) the CI

$(11.09;\ 13.71)$ contains 12 — it does not. The CI is more informative: it shows not only "yes/no" but also the size of the effect.

6 Significance and practical importance

With a huge sample a negligible effect becomes "statistically significant": $\bar{x} = 30.3$ against $\mu_0 = 30$ with $\sigma = 4$ and $n = 10\,000$ gives $z = 7.5$, $p < 10^{-12}$, but a difference of 0.3 (1%) may have no practical meaning. And conversely: a large effect in a small sample may be "insignificant" (low power). Always report the effect size and CI along with p.

If many hypotheses are tested, then at $\alpha = 0.05$ about 5% of true nulls will be "significant" by chance. 20 independent tests give a chance of at least one false alarm of $1 - 0.95^{20} = 64\%$. For such cases corrections are applied (for example, Bonferroni: α/m).

7 Key terms

Term	*In Russian*	Meaning
Null hypothesis	*Нулевая гипотеза*	H_0 — "no effect"
Alternative hypothesis	*Альтернативная*	H_1 — what we want to prove
Test statistic	*Статистика критерия*	z or t
p-value	*p-значение*	$P(\text{more extreme} \mid H_0)$
Significance level	*Уровень значимости*	$\alpha = P(\text{Type I error})$
Type II error	*Ошибка II рода*	β
Power	*Мощность*	$1 - \beta$

Formulas of Lecture 10

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}, \qquad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}\ (\nu = n - 1), \qquad z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}};$$

p-value: $2P(Z > |z|)$ or $P(Z > z)$ or $P(Z < z)$;

decision: $p \le \alpha \Rightarrow$ reject H_0;

$$\text{power} = 1 - \beta; \qquad n \approx \left(\frac{(z_{\alpha/2} + z_\beta)\sigma}{\Delta}\right)^2.$$

8 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Formulate H_0 and H_1 (stating the type of test): (a) a manufacturer claims the average pack mass is 500 g, an inspector suspects underweight; (b) a bank wants to check whether the share of overdue payments has changed compared with the previous 6%; (c) a manager believes that a new application form has shortened the average processing time from 15 minutes.

Problem 2 [A] $H_0 : \mu = 50$, $H_1 : \mu \neq 50$, $\sigma = 10$, sample $n = 100$, $\bar{x} = 52.5$, $\alpha = 0.05$. Carry out the test.

Problem 3 [A] A study obtained $p = 0.03$. Which statements are true? (1) The probability that H_0 is true is 3%. (2) If H_0 were true, a result at least as extreme as the observed one would occur in 3% of samples. (3) The effect is large. (4) At $\alpha = 0.05$ H_0 is rejected; at $\alpha = 0.01$ it is not.

Problem 4 [B] $H_0 : \mu = 50$, $H_1 : \mu \neq 50$. A sample $n = 9$, $\bar{x} = 52$, $s = 6$. Carry out the t-test ($\alpha = 0.05$) and find the p-value.

Problem 5 [B] The average delivery time should be 250 minutes. For 16 orders $\bar{x} = 245$, $s = 20$. Test $H_0 : \mu = 250$ against $H_1 : \mu < 250$.

Problem 6 [B] A company claims that 30% of customers use the app. In a random sample of 500 customers, 170 use the app.

(a) Test $H_0 : p = 0.30$ against $H_1 : p \neq 0.30$ at $\alpha = 0.05$.
(b) How does the conclusion change for $H_1 : p > 0.30$?
(c) Construct a 95% CI and compare with the result of (a).

Problem 7 [B] Describe Type I and Type II errors in the language of the problem: (a) a bank’s monitoring system tests H_0: “the transaction is normal”; (b) testing a new drug, H_0: “the drug has no effect”. Which error is more costly in each case and how does this affect the choice of α?

Problem 8 [B] $H_0 : \mu = 100$, $\sigma = 15$, $n = 36$, $\alpha = 0.05$ (two-sided).

(a) Find the power of the test at a true $\mu = 105$.
(b) What n is needed for 80% power at $\mu = 105$?

Problem 9 [B] $n = 64$, $\bar{x} = 103$, $\sigma = 16$. (a) Construct a 95% CI for μ. (b) Without conducting a test, determine the result of a two-sided test at the 0.05 level for $H_0 : \mu = 100$ and for $H_0 : \mu = 98$. (c) Verify this by computing z and p.

Problem 10 [B] An analyst tested 10 independent hypotheses at the level $\alpha = 0.05$ and found one “significant”. (a) What is the probability of at least one false alarm if all 10 null hypotheses are true? (b) What significance level for each test does the Bonferroni correction give?

Problem 11 [C] Show that when H_0 is true and the statistic is continuous, the p-value (for a right-sided test) is uniformly distributed on $[0,1]$, i.e. $P(p \le \alpha) = \alpha$ for any α.

Problem 12 [C] Show that a two-sided z-test at level α rejects $H_0 : \mu = \mu_0$ if and only if μ_0 does not belong to the $(1-\alpha)$ CI $\bar{x} \pm z_{\alpha/2}\sigma/\sqrt{n}$. Then, for $n = 10\,000$, $\bar{x} = 30.3$, $\sigma = 4$, $\mu_0 = 30$, find the p-value and the CI and comment on the practical significance.